

Student Performance Prediction & Academic Analysis System

AI/ML - Week 1 Corporate Project
Project ID: AIML-W1-001

Domain: Artificial Intelligence | Machine Learning | Python | Data Analysis

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Date: 07-10-2026

1. Introduction and Problem Statement

An educational institute wants to identify students who are at risk of failing so that they can receive additional academic support before the final examination. The institute records study hours, attendance, previous marks, assignment scores, internal marks and the number of previous failures for every student.

2. Objective

To build a simple Machine Learning system that predicts whether a student will **Pass** or **Fail** using this information, and to convert the prediction into a simple recommendation for teachers. The project also introduces the basic ML workflow: loading data, cleaning, analysis, visualisation, model training, prediction and evaluation.

3. Technologies Used

Technology	Purpose
Python	Programming language
Jupyter Notebook	Development environment
Pandas	Data handling and analysis
NumPy	Numerical operations
Matplotlib	Data visualisation
Scikit-learn	Machine Learning model and evaluation
CSV	Dataset format

4. Dataset Description

The dataset *Student_Performance.csv* is synthetic (generated with Python) and contains 202 raw rows, which became 200 student records after cleaning. Each row describes one student.

Column	Description
Student_ID	Unique student ID
Study_Hours	Average daily study hours
Attendance	Attendance percentage
Previous_Marks	Previous examination marks
Assignment_Score	Assignment marks

Internal_Marks	Internal examination marks
Previous_Failures	Number of previous failures
Final_Result	Target: Pass / Fail

5. Project Workflow

Dataset -> Data Understanding -> Data Cleaning -> Data Analysis -> Visualisation -> Feature Selection -> Train/Test Split -> ML Model -> Prediction -> Evaluation -> Final Insights

6. Data Cleaning

Checking data quality showed 5 missing values (3 in Attendance, 2 in Assignment_Score) and 2 duplicate rows. Missing values were filled with the median of their column, because the median is not affected by extreme values. Duplicate rows were removed. After cleaning, there were no missing values and no duplicates, leaving 200 students.

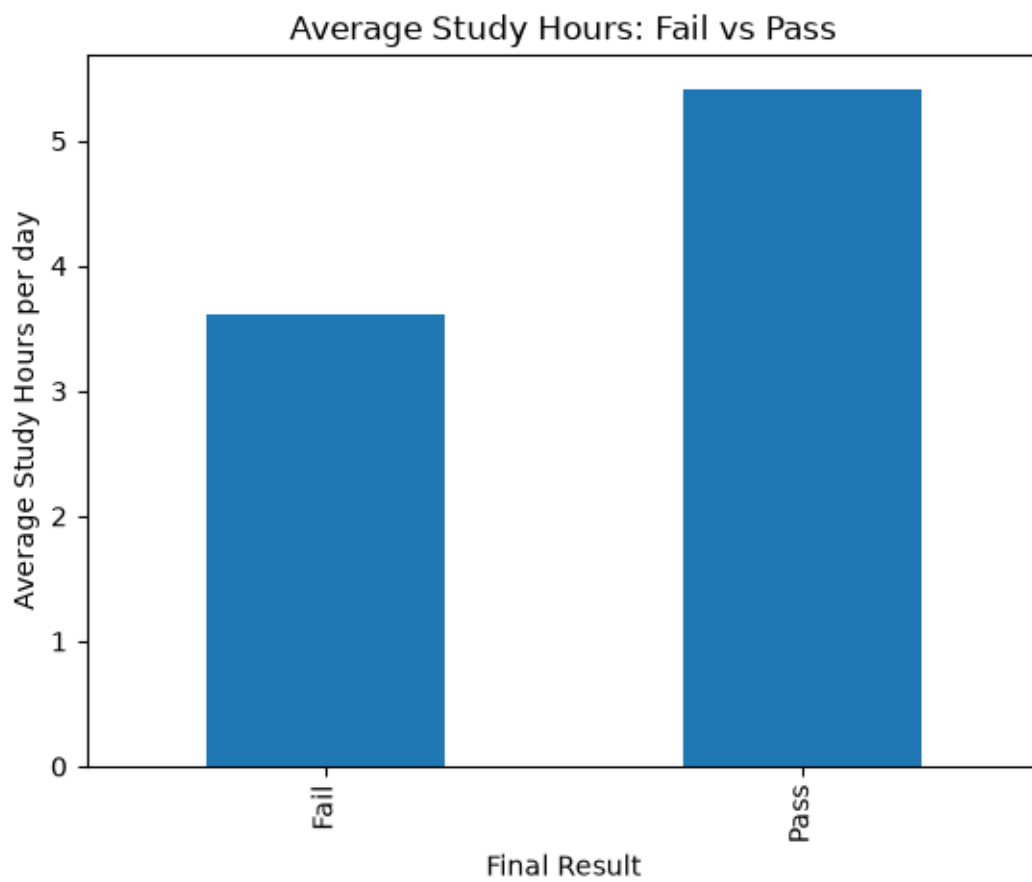
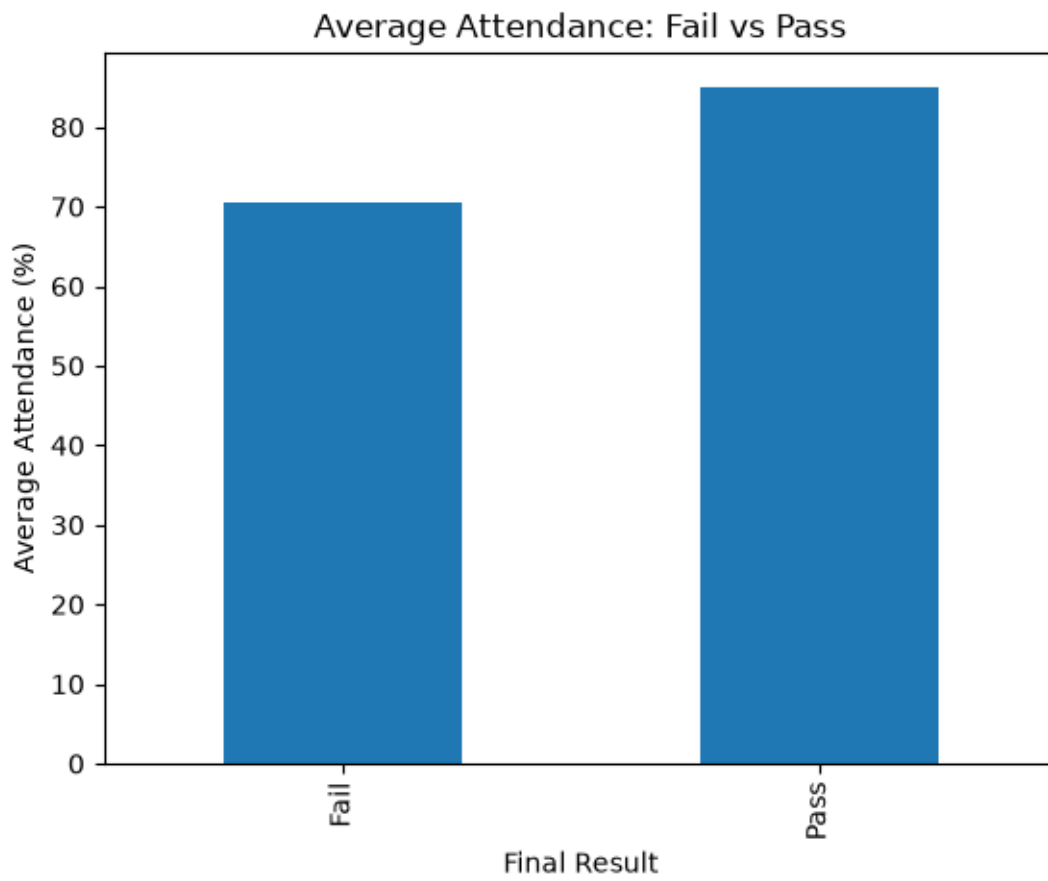
7. Data Analysis

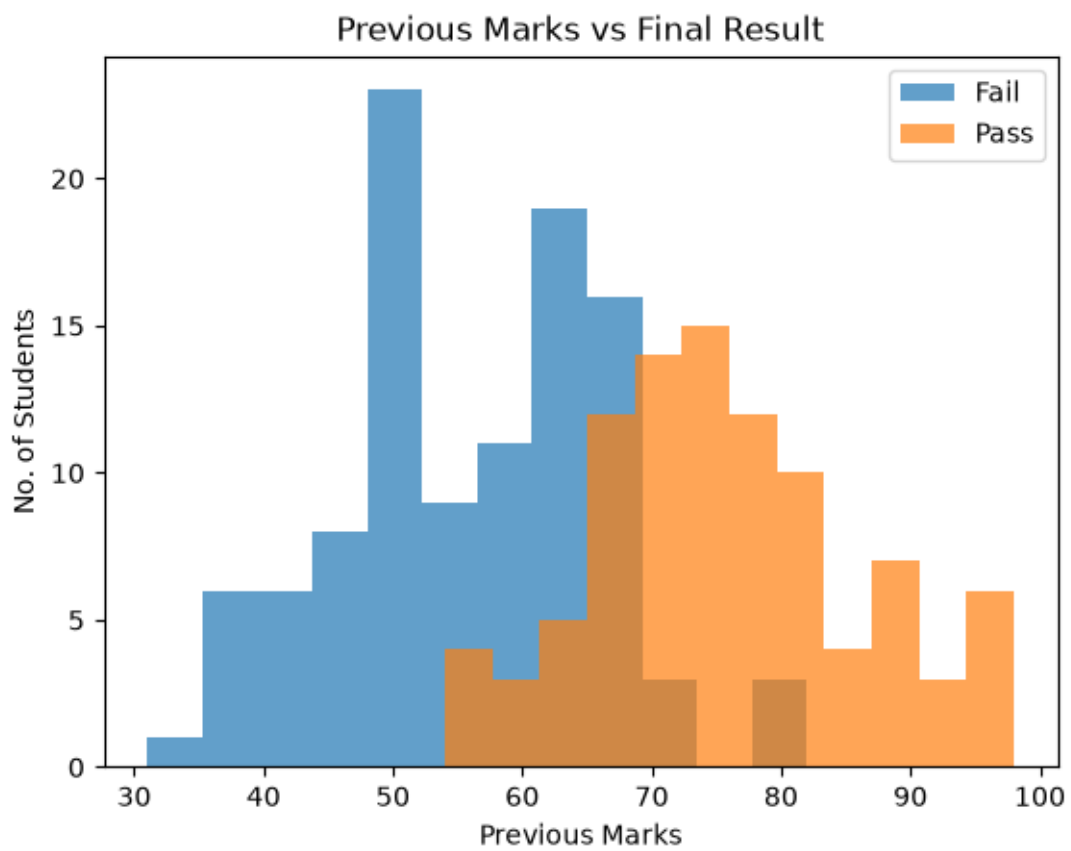
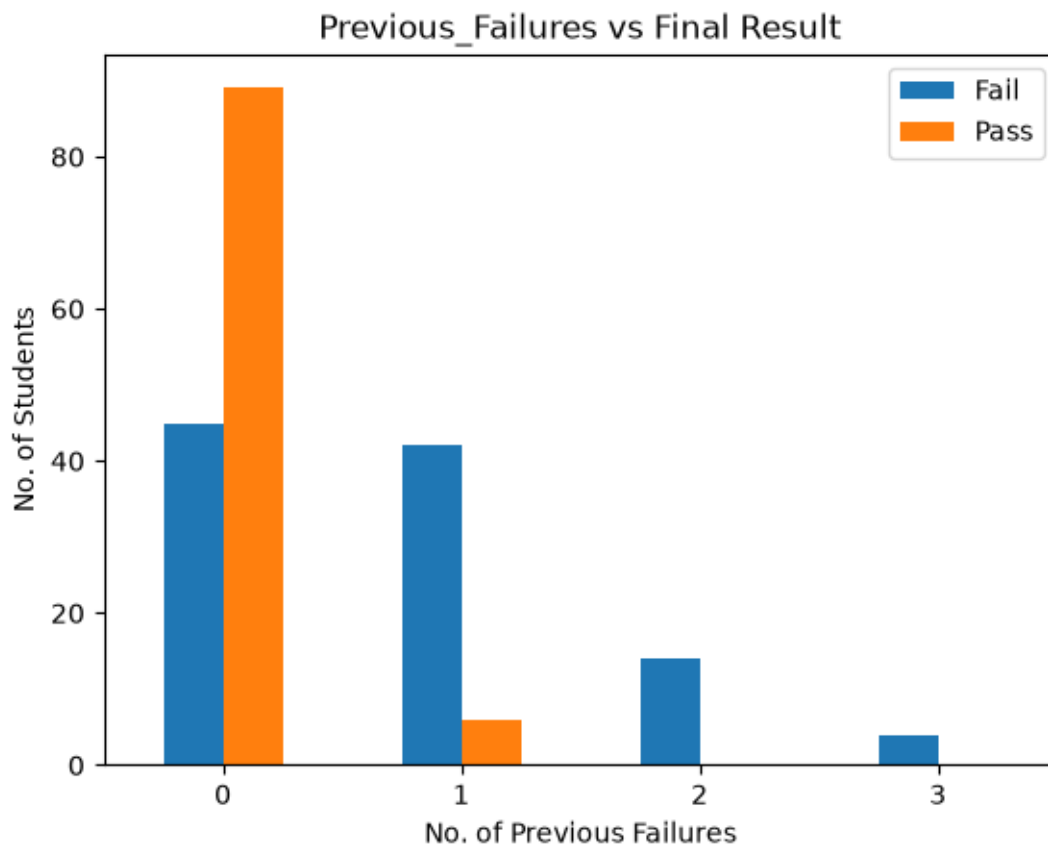
Students were grouped by final result and the average of each feature was compared:

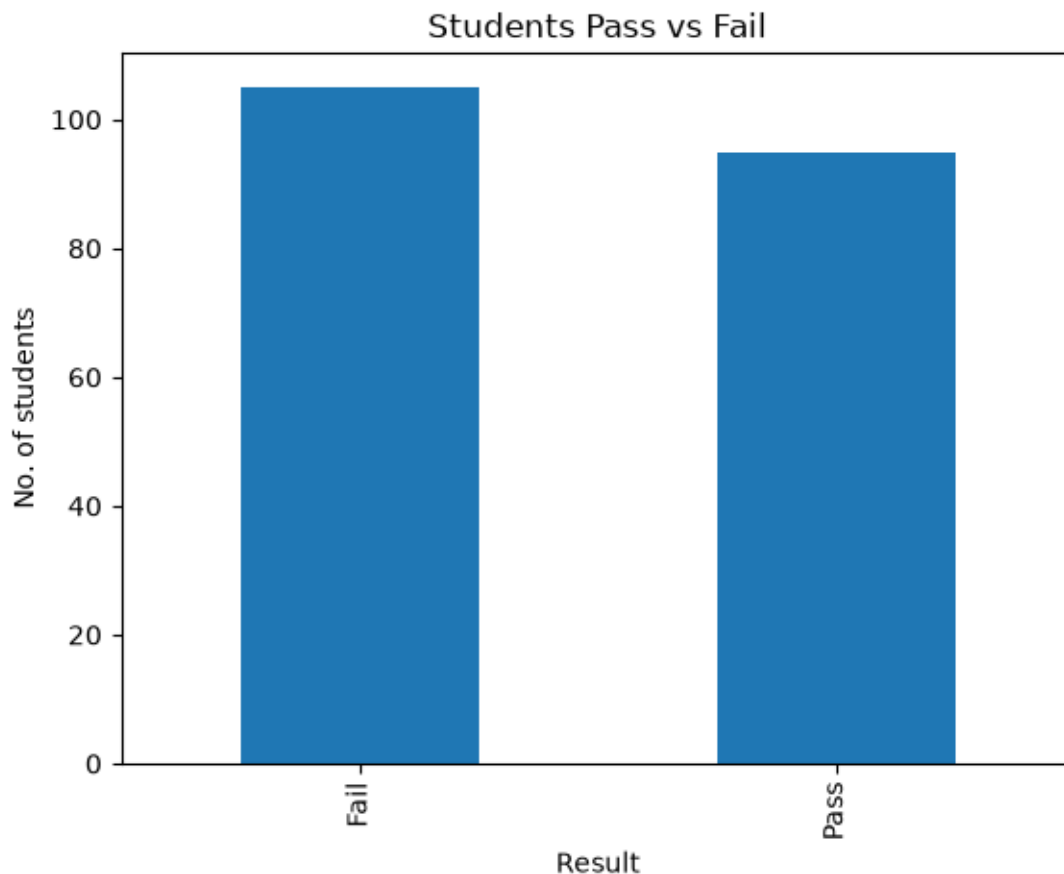
Feature	Fail (average)	Pass (average)
Study hours per day	3.6	5.4
Attendance (%)	70.4	85.0
Previous marks	55.8	75.6
Previous failures	0.78	0.06

Answers to the analysis questions: (1) higher attendance goes with better results; (2) students who study more hours generally pass; (3) previous marks strongly affect the final result; (4) students with previous failures are much more likely to fail; (5) Previous_Marks has the strongest relationship with the result (correlation 0.70), while Previous_Failures is negatively related (-0.50).

8. Data Visualisation







9. Machine Learning Model

The target Final_Result was converted to numbers (Pass = 1, Fail = 0). The six input columns were used as features; Student_ID was excluded because it carries no information. The data was split 80% / 20% into 160 training and 40 testing students (random_state = 42), so the model could be tested on students it had never seen.

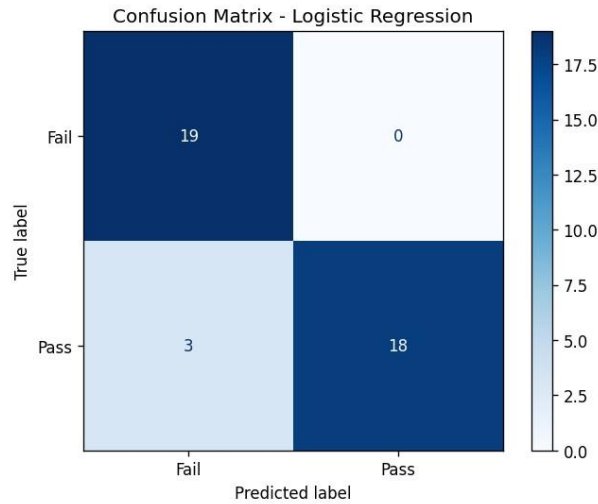
Logistic Regression was used as the main model. It assigns a weight to each feature, converts the weighted sum to a probability of passing, and predicts Pass when that probability is at least 0.5. As an optional second model a **Decision Tree** was also trained on the same split.

10. Results and Evaluation

Model	Training accuracy	Testing accuracy
Logistic Regression	94.4%	92.5%
Decision Tree	100.0%	87.5%

Logistic Regression performed better on unseen students. The Decision Tree scored 100% on training data but lower on test data, which is a sign of overfitting (memorising the training data).

The confusion matrix of the Logistic Regression model shows 19 correct Fail predictions, 18 correct Pass predictions, 0 students wrongly predicted as Pass and 3 students wrongly predicted as Fail.



11. Prediction System and Recommendations

A function accepts the six details of a student, predicts Pass or Fail, and prints a recommendation: for Pass, "Performance is satisfactory. Continue current study pattern"; for Fail, "Student requires additional academic support. Improve attendance and study hours".

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STUDENT PERFORMANCE PREDICTION
=====
Study Hours      : 3
Attendance       : 68
Previous Marks   : 52
Assignment Score : 60
Internal Marks   : 55
Previous Failures : 1

Predicted Result : FAIL
Chance of passing : 0.0%

Recommendation:
Student requires additional academic support.
Improve attendance and study hours.
=====

=====
STUDENT PERFORMANCE PREDICTION
=====
Study Hours      : 7
Attendance       : 94
Previous Marks   : 85
Assignment Score : 90
Internal Marks   : 88
Previous Failures : 0

Predicted Result : PASS
Chance of passing : 100.0%

Recommendation:
Student performance is satisfactory.
Continue current study pattern.
=====
```

12. Final Insights

- Of 200 students, 95 passed (47.5%) and 105 failed (52.5%).
- Passing students attend about 85% of classes versus 70% for failing students.
- Passing students study about 5.4 hours per day versus 3.6 hours.
- Previous failures are strongly linked to failing again.
- The model predicts Pass/Fail with about 92.5% accuracy on unseen students.

13. Conclusion

The project shows that a simple Machine Learning model can use academic and attendance data to identify students at risk of failing. The institute can use these predictions to give early support to the students who need it most. Because the dataset is synthetic, real-world accuracy would likely be lower, but the workflow would remain the same.

14. Future Scope

- Use real, anonymised student data.
- Add more features such as sleep, health, family support and class participation.
- Try other models (Random Forest) and cross-validation.
- Predict actual marks instead of only Pass/Fail.
- Build a web application for teachers.